

INTRO ANIMATION – ONE PAGE OVERVIEW

1. Component Structure

App.js

- PerfectIntroAnimation.js
- ParticleGlobe.js
- LogoAndText.js
- CameraController.js

2. How It Works

- App.js loads PerfectIntroAnimation.
- PerfectIntroAnimation creates the overlay, blur, fade-out timing, and 3D canvas.
- Inside the Canvas:
 - ParticleGlobe creates and animates 6000+ particles into a sphere.
 - LogoAndText fades the logo in, holds it, and fades it out.
 - CameraController moves the camera from $z = 3.2$ to $z = 7.9$ smoothly.

3. Animation Timeline

- Gather Duration: 3.6s
- Hold Duration: 0.8s
- Vanish Duration: 1.6s
- Logo stays visible longer by delaying fade-out.

4. Why Camera Starts Close ($z = 3.2$)

- Particles begin far away (6–20 units).
- Starting close gives a cinematic explosion feel.
- Camera slowly pulls back to reveal the full sphere.

5. Particle Design

- Each particle uses a glowing circular texture created in effects.js.
- Colors: #ffd97b, #ffc54d, #fff3c4, #ffffff
- Additive blending for natural glow.
- Wave motion keeps particles alive.

6. Logo Glow

- Custom radial gradient texture.
- Transparent edges to avoid box glow.
- Glow expands and contracts smoothly.

7. File Purposes

- ParticleGlobe.js = Handles particles, movement, rotation, vanish.
- LogoAndText.js = Handles logo fade-in, glow, fade-out.
- CameraController.js = Smooth camera movement.
- PerfectIntroAnimation.js = Layout, timing, blur overlay.
- effects.js = Shared glow/texture helpers and easing functions.

